License Plate Reading Research Plan

By Christopher Paquin

The goal of this research is to identify and compare several ways to read plates from cycling incident photos and plan small demo projects. I will focus on existing plausible approaches and use AI assistance to build the testing demos to save time while attempting to provide proof of concept. Using AI in this way was approved and even suggested by our project sponsor and cleared with the instructor.

There are 4 approaches I want to investigate for various reasons listed below.

- Google ML Kit
- Plate Recognizer Snapshot Cloud
- Gemini API
- FastALPR

Google ML Kit: I chose this to explore whether license plates can be read directly on a phone without an internet connection or a per-image fee. Through the demo, I hope to learn how accurately it reads cropped plates, how quickly it runs on a real phone, and how much additional work is needed to distinguish plate numbers from other text.

Plate Recognizer Snapshot Cloud: I chose this because it is designed specifically to locate and read license plates, providing a useful baseline for comparison. Through the demo, I hope to learn how well a dedicated service handles our incident photos, how easily it connects to our app, and whether its accuracy and response time justify the subscription cost.

Gemini: I chose this AI to explore whether a general-purpose AI model can read license plates with minimal specialized development. Through the demo, I hope to learn how it handles unclear images, whether it reliably identifies unreadable plates instead of guessing, and how its accuracy, processing time, and cost compare with dedicated recognition tools.

FastALPR: I chose this to explore a customizable solution running on an AWS server I own. Through the demo, I hope to learn how well it perform on our images, what computing resources they require, and how much effort it takes to build a service that receives photos and returns plate values to the app.

By making these demos, I hope to measure how often each approach reads the entire license plate correctly, returns an incorrect value, or identifies a plate as unreadable. I also

want to compare processing time, cost per image, and resource use, including phone battery consumption or server requirements. Testing the same images across all four approaches will help me understand how image quality affects performance and which option offers the best balance of accuracy, speed, cost, and ease of integration.